



Claudia Torres Carretón

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Profile

Physics graduate with a minor in Computing and Data Analysis, specializing in solving high-complexity technical problems. My differential value lies in the ability to bridge the rigor of classical scientific methodology with cutting-edge technology. Beyond analysis, my focus is on strategic implementation and continuous improvement, constantly seeking to integrate disruptive advancements to scale solutions that agilely address today's challenges. I possess proven resilience in high-pressure environments and a constant drive to contribute to projects with high-impact potential.

Education

Bachelor's Degree in Physics

Sep 2022 - Jun 2026

Universidad Europea, Valencia, Valencia

Minor in Computing & Data Analysis.

Relevant subjects and technical skills:

Physics & Scientific Computing

- Optics: Design & simulation (Photopia, DIALux), Photon-based systems.
- Quantum Tech: Algorithms (Deutsch-Jozsa, Grover, Shor), Cryptography (BB84, E91), System simulation.
- Scientific Modeling: Complex physical systems & Signal/Image processing.
- Languages: Python, C++, Mathematica.

Advanced Analytics (Exploratory Data Analysis, Machine Learning) & Artificial Intelligence

- Core ML: Supervised/Unsupervised, Neural Networks, Evolutionary Computing.
- Deep Learning & NLP: TensorFlow, PyTorch, Transformers, Word Embeddings.
- Statistical Math: Bayesian Analysis, Time Series, Dimensionality Reduction (PCA, SVD, L1/Lasso).
- Model Governance: Validation (Bias-Variance), Confusion Matrix, EM Algorithms.

Fundamentals of Big Data

- Data Engineering: End-to-end Pipelines (ETL), Data Architecture & Lifecycle.
- Infrastructure: Distributed Computing (Hadoop, PySpark), Cloud Solutions.
- Databases: SQL & NoSQL Management.
- DevOps: Docker, Git/GitHub, AI Infrastructure Deployment.

Personal details

Date of birth

January 28, 2004

Driver's license

B

Languages

Spanish (Native)

Valencian (Native)

English (B2)

Note: Strong background in English-mediated education. Both my secondary education (Scientific Track) and several advanced modules during my Bachelor of Science in Physics were conducted in English, covering core STEM subjects and technical scientific documentation.

French (A1)

Hobbies

- Practicing any sport.
- Reading books, especially about popular science.
- Traveling, learning about cultures and languages.
- Attend events where you can grow personally and academically.

Grade Average: 8.25
Grade Average (GPA): 3.3

Bachelor's Thesis

Sep 2025 - Present

Universidad Europea

Title: Quantum Proton Tunneling in DNA: Open Quantum Systems Simulation, Machine Learning Inference, and Quantum Computing Proof-of-Concept.

Supervised: Hector Gisbert Mullor

Description:

- Applied Stochastic Dynamics and numerical simulations to model complex biological environments.
- Leveraged Machine Learning for inverse problem solving and environmental parameter inference.
- Analyzed high-dimensional systems using advanced computational physics and explored Quantum Computing proof-of-concepts.

Certificates

Team leadership

2024

Harvard ManageMentor Course.

Volunteering

Basic Robotics Instructor

2024

CEIP José María Oltra, Montcada, Valencia

Throughout the year, I taught a total of three different classes to third and sixth grade students on Micro:bit, with the aim of teaching them its structure and basic operation. (Total: 18 hours)

Cybersecurity Startup Hackathon

Feb 2025

INCIBE and Startup Valencia, Poblats Marítims, Valencia

Intensive search for solutions to challenges proposed by Telefónica Tech, Fourvenues, and Internxt. (Total: 2 days)

NASA Space Apps

Oct 2025

Universidad Politécnica de Valencia, Valencia

Use of machine learning for the search for exoplanets. (Total: 3 days)

Work experience

Waitress

2022 - 2025

La Tagliatella, Valencia

During the summers of this period. (Total: 9 months)